

|                   | Copy-on-Write (CoW)                                    | Merge-on-Read (MoR)                              |
|-------------------|--------------------------------------------------------|--------------------------------------------------|
| Write Performance | ✗ <b>Slow</b> - rewrites entire file                   | ✓ <b>Fast</b> - writes delete files + changes    |
| Read Performance  | ✓ <b>Fast</b> - direct file reads                      | ✗ <b>Slow</b> - merges delta at read time        |
| Storage Overhead  | ✗ <b>High</b> - duplicates data on updates             | ✓ <b>Low</b> - stores only changes               |
| Table Maintenance | ✓ <b>Low</b> - files already optimized                 | ✗ <b>High</b> - needs frequent compaction        |
| Best For          | <b>Read-heavy workloads</b><br>Analytics (BI, reports) | <b>Write-heavy workloads</b><br>(Streaming, CDC) |